

Measuring cybercrime: data sources and challenges

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CRIME DATA SOURCES: AN OVERVIEW

Traditional crimes can be measured through three main data sources:

- Official crime statistics
- Non official statistics (victimisation and self-report surveys)

What about cybercrime? It is very difficult to measure it and to reconcile data from different sources!

OFFICIAL CRIME STATISTICS

- Official crime statistics = quantitative data on crime recorded by state agencies, primarily police, prosecutors, and courts
- Originated in the early 19th century in France
- Initially focused on judicial and penitentiary statistics, later also on police data
- Main administrative sources at the supranational level: Interpol (until 2006), UN (CTS), European Sourcebook, Eurostat database
- Key problems: the so called dark figure of crime + no detailed info on the criminal event, the victim and the author

OFFICIAL CRIME STATISTICS AND CYBERCRIME

- Official statistics on cybercrime are extremely limited
- cybercrimes (especially cyber-enabled) are in most cases registered as traditional crimes
- lack of uniform registration of cybercrime in the police registration systems of many western countries
- Exceptions:
 - FBI Internet Crime Complaint Center <https://www.ic3.gov/>
 - Uniform Crime Report Survey of Canada
 - European Sourcebook of Crime and Criminal Justice Statistics publishes cross-national records of cyber-enabled fraud, and for some countries cyber-dependent crime

OFFICIAL CRIME STATISTICS AND CYBERCRIME

- Information concerning different types of cybercrime may be present in the qualitative, unstructured, text fields found in the police/court registration system
- ... but retrieving this information from unstructured text fields is a labour-intensive task!

OFFICIAL CRIME STATISTICS AND CYBERCRIME

- To overcome this issue....

Using Machine Learning to Detect ICT in Criminal Court Cases

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Text Mining for Cybercrime in Registrations of the Dutch Police

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VICTIMISATION SURVEYS

Definition:

a technique that involves asking a representative sample of the population:

- whether and of what crimes they have been victims in a given period of time prior to the interview...
- and, if so, further details about the incident (e.g., where it occurred, at what time, and how).

THE US NCVS

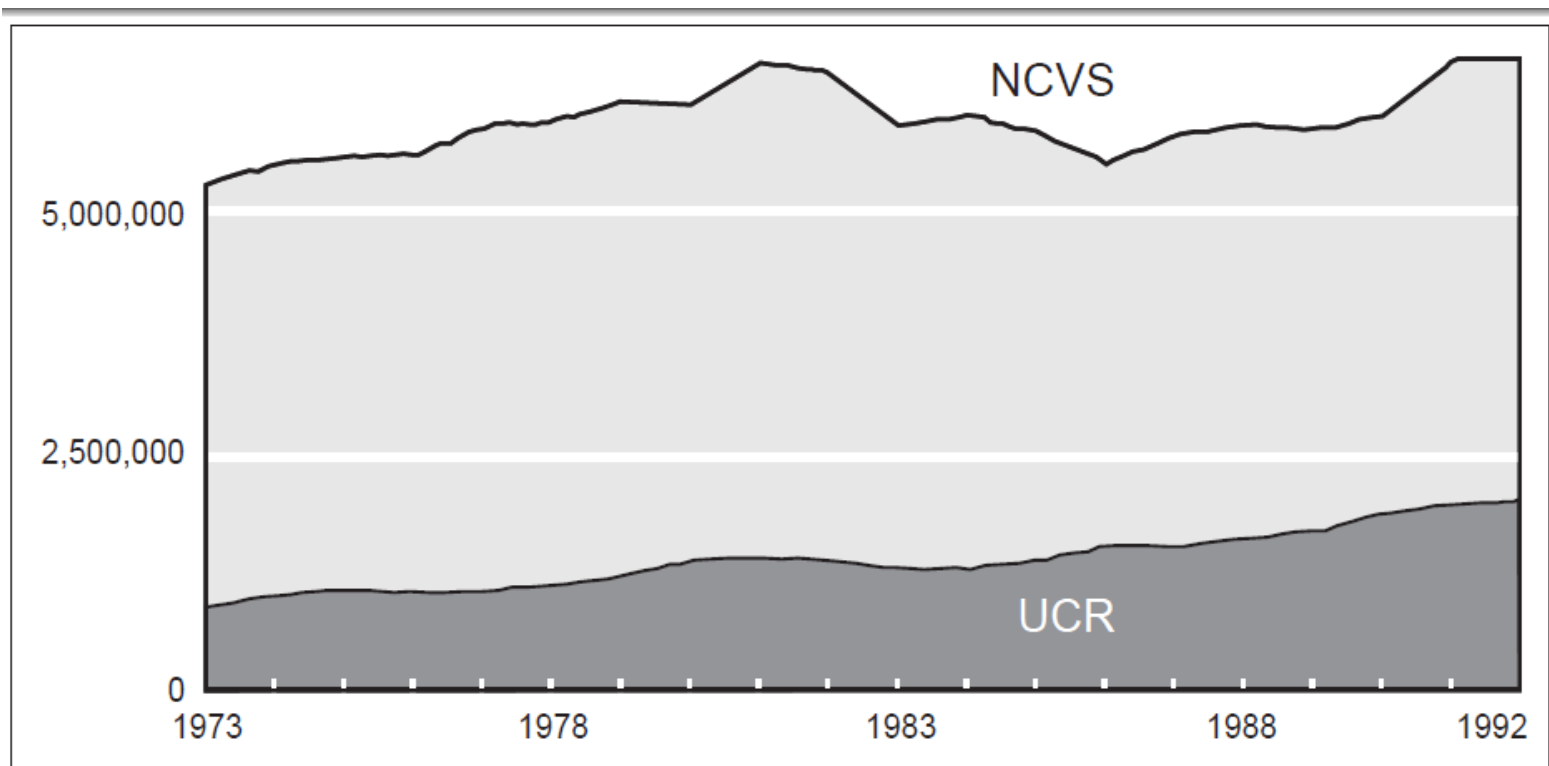
- First pilot studies in the 1960s on 10,000 people: respondents reported DOUBLE the crime rate recorded!
- Since 1972, the National Crime Victimization Survey (NCVS) has been investigating the frequency, characteristics, and consequences of victimization.
- Focusing on certain crimes against the person (rape and sexual assault; robbery; assault; bag snatching/pickpocketing)... & against property (theft; burglary; motor vehicle theft).

THE US NCVS

It allows us to:

- 1) estimate the total number of crimes committed against the population as a whole

Figure 1 – Violent crime in the United States. Comparison of NCVS and UCR. Years 1973–1992. Absolute values.



Source: Zawitz et al. (1993).

THE US NCVS

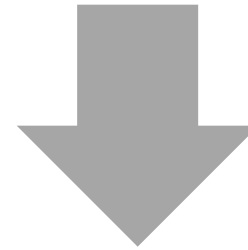
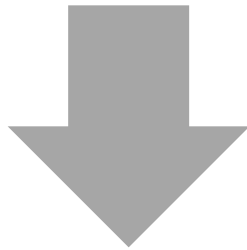
2) Investigate the levels of victimization to which different segments of the population (women, the elderly, etc.) are exposed, based on various information about the victim (e.g., age, gender, ethnicity, marital status, income, education) collected.

E.g., males/blacks/young people up to age 24 are more at risk of violent crime than women/whites/over 24s.

3) Provide details about the perpetrator and the events that occurred.

THE IMPACT OF THE US NCVS

HUGE IMPACT of the NCVS.
Similar surveys developed...



... at the international level / ... at the national level

ICVS

es.: NL, Victims of Crime, '74
UK, British Crime Survey, '82
IT, '97

VICTIMISATION SURVEYS AND CYBERCRIME

- What about cybercrime?

Reep-van den Bergh and Junger *Crime Sci* (2018) 7:5
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 Crime Science

RESEARCH

Open Access



Victims of cybercrime in Europe: a review of victim surveys

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VICTIMISATION SURVEYS AND CYBERCRIME

- Objective: To map cyber-victimization patterns in Europe, to provide estimates of annual prevalence rates (number of victims per population) for six cybercrimes.
- Methodology: Definition of survey inclusion criteria; database and internet searches, grey literature, and contacts with national statistical offices.
- Results: Nine surveys identified; different annual prevalence rates depending on the crime: e.g., 1-3% for online shopping scams; 3% of the population is a victim of cyberbullying.

SELF-REPORT SURVEYS AND CYBERCRIME

- surveys in which respondents are asked about instances in which they have been actively involved in crime or deviant behavior
- key for measuring juvenile delinquency, and developing and testing theories of criminal and deviant behavior
- Key survey: ISRD <https://isrdstudy.org/>

SELF-REPORT SURVEYS AND CYBERCRIME

- On cybercrime:

Name	Acronym	Country	Frequency	Round analyzed	Sample
Offending, Crime and Justice Survey	OCJS	UK	Annual, 2003 to 2006	2006	5,354 respondents aged 10 to 29
Youth Delinquency Survey	YDS	Netherlands	Quinquennial	2015	1,471 respondents aged 12 to 23
International Self-Report Delinquency Study	ISRD4	Cross-national	1991/91, 2006/08, 2012/19, 2021/22	2021/22	Varies by country (respondents aged 13 to 17)

Source: Buil-Gil, Trajtenberg, Aebi (2025)

SELF-REPORT SURVEYS AND CYBERCRIME

*Table 6. Measures of cybercrime offending included in self-report crime surveys**

	<i>Cyber-dependent crime</i>				<i>Cyber-enabled financial crime</i>		<i>Cyber-enabled personal crime/deviance</i>	
	Malware	Hacking	DoS	Digital piracy	Online banking fraud	Online shopping fraud	Hate speech	Online harassment
OCJS	"used internet to send viruses on purpose to other computers"	"used internet to hack into other computers? by hacking we mean using a computer to illegally get access to another computer's files"		"used internet to download software, music or films that you knew to be pirated or unauthorized"	"bought anything over the internet using payment card or card details that did not belong to you, without the card owner's permission"			"sent email message to someone in order to harass, scare or threaten them"; "sent voice or text message on your phone..."
YDS	"intentionally sent out viruses through e-mail or over the internet"	"logged on to somebody else's computer, email or social media account without informed consent"; "changed someone's account password (computer or social media) to prohibit them from accessing"; "logged on onto somebody else's computer, email or social media account without informed consent, and manipulated or deleted information"	"tried to disrupt a website or email account by sending out large amounts of data"			"sold something through internet, but not sending out goods after receiving payment"; "bought something through internet, but not paying for goods after receiving items"		"threatened someone through text messages, e-mails or in chat boxes"; "threatened someone through social media, such as WhatsApp, Facebook, Twitter, etc."
ISRD4		"hacked or broken into a private account or computer to acquire data, get control of an account, or destroy data"				"used internet, e-mail or social media to dupe or deceive others (like phishing, selling worthless or illegal things, etc.) to make money"	"sent hurtful messages or comments on social media about someone's race, ethnicity or nationality, religion, gender identity, etc."	

Source: Buil-Gil, Trajtenberg, Aebi (2025)